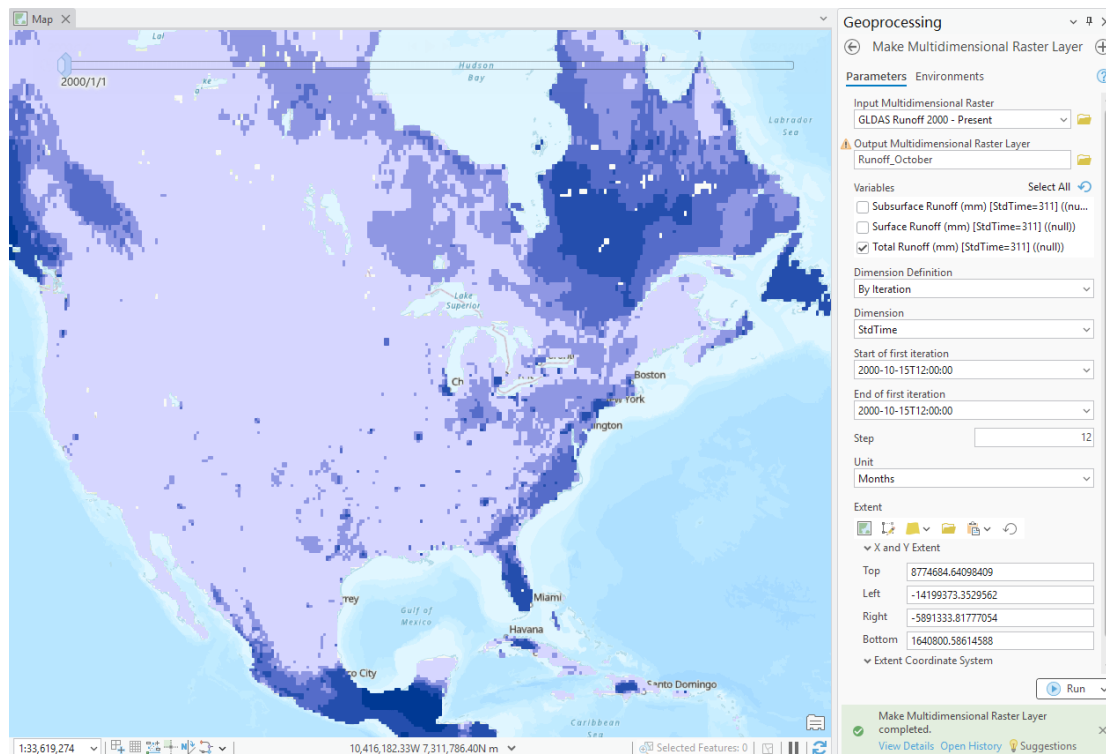


GGR276 Assignment 2 Part A

Q1

Viewing October precipitation in different years across the United States using the time slider reveals significant differences in October precipitation. For example, October 15th, 2004, saw relatively high precipitation, while October 15th, 2007, saw almost no precipitation. Furthermore, the spatial distribution of precipitation also shows certain patterns, with precipitation concentrated almost entirely on the east and west coasts and near the Gulf of Mexico, while the central region receives almost no precipitation. For instance, the New York area receives precipitation almost every October, while the vast central region receives almost no precipitation each year.



Q2

Several spatial distribution patterns can be observed on the map. The highest average precipitation values are found in the Northwest Pacific and parts of the eastern United States, indicating consistently high precipitation in October in these areas. Coastal areas typically experience higher precipitation due to the influence of moist air masses from nearby oceans. In contrast, the southwestern and western inland parts of the

United States have lower average precipitation, and these areas generally have drier climates.

Areas with higher standard deviations indicate greater interannual variability in precipitation. The map shows significant precipitation fluctuations in parts of the eastern United States, the Gulf Coast, and some coastal areas. These areas may experience significant precipitation fluctuations due to the influence of tropical storms, hurricanes, and other large weather systems, which bring heavy rainfall in some years and not in others. In contrast, the standard deviation is lower in most of the western and southwestern inland parts of the United States, indicating more stable interannual variability in October precipitation in these areas.

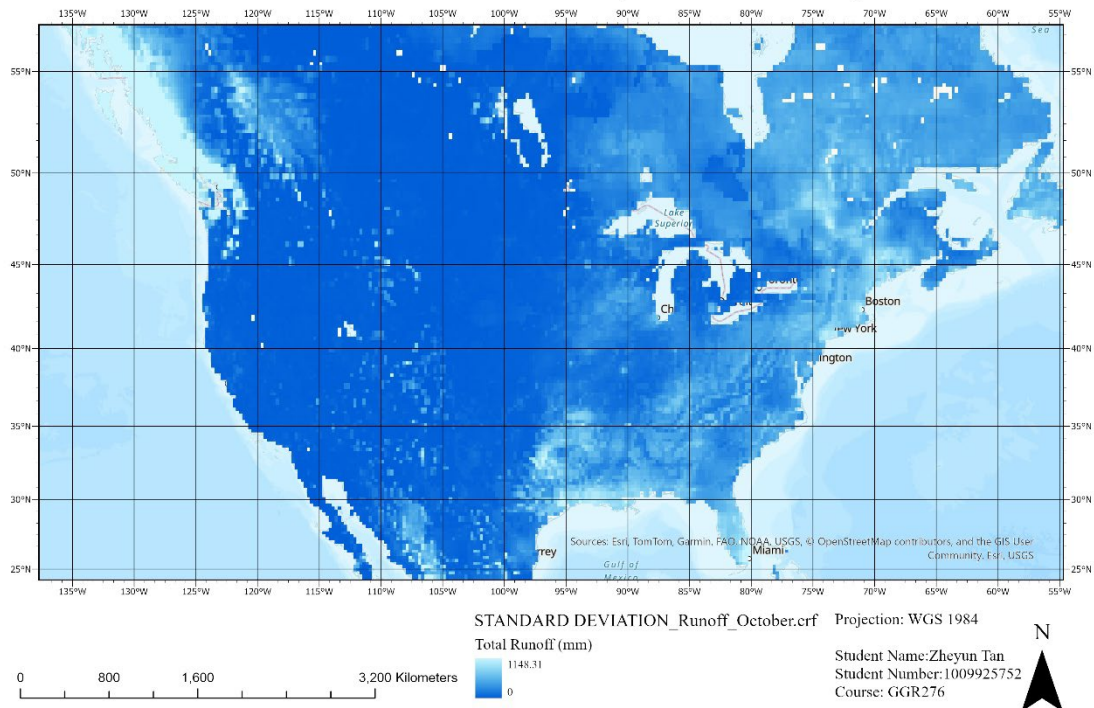
Minimum values represent the driest October precipitation at each location during the study period. In many parts of the western and southwestern inland United States, minimum precipitation is near zero, and these areas receive almost no rainfall in October in some years, indicating arid or semi-arid climates. In contrast, some coastal areas and parts of the eastern United States have slightly higher minimum precipitation. Even in the driest years, these areas tend to receive some rainfall, likely due to the influence of nearby moisture sources.

Maximum precipitation values show the wettest weather conditions experienced by locations in October during the study period. According to the map, maximum precipitation is higher along the Gulf Coast, in the southeastern United States, and in parts of the Pacific Northwest. These areas are more susceptible to tropical storms, hurricanes, or strong coastal weather systems, resulting in extreme precipitation events. In contrast, maximum precipitation is lower in many parts of the western inland United States and parts of the southwestern United States. Even in the wettest years, these areas receive less rainfall than the coastal and eastern regions.

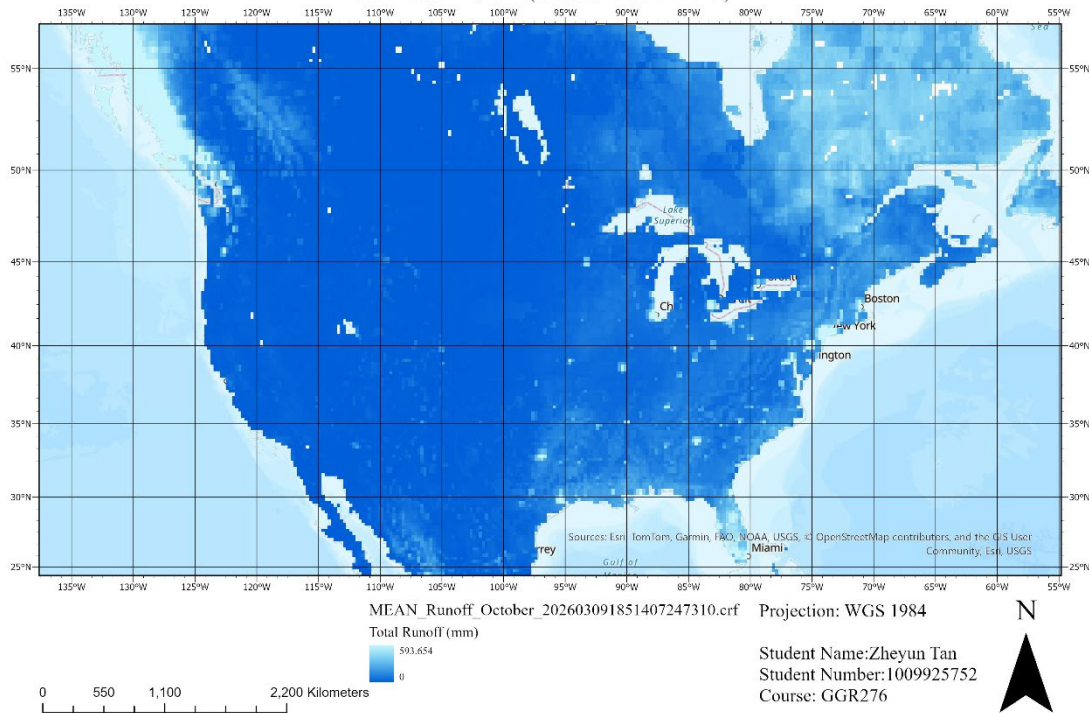
The large range of precipitation variation indicates significant differences in precipitation between the driest and wettest years in these regions. Rainfall variation is significant in parts of the southeastern United States, the Gulf Coast, and some coastal areas, indicating that October rainfall in these regions can range from relatively dry to extremely wet. This variation may be influenced by tropical storms, hurricanes, and other severe weather systems that occasionally bring heavy rainfall. In contrast, rainfall variation is relatively small in many parts of the western interior and southwestern United States, indicating that October rainfall in these areas is lower than annually. The climate in these regions is generally more stable but also drier overall, resulting in smaller differences between minimum and maximum rainfall.

Q3

Standard Deviation of Total Runoff (October 2000–2025)



Mean Total Runoff (October 2000–2025)

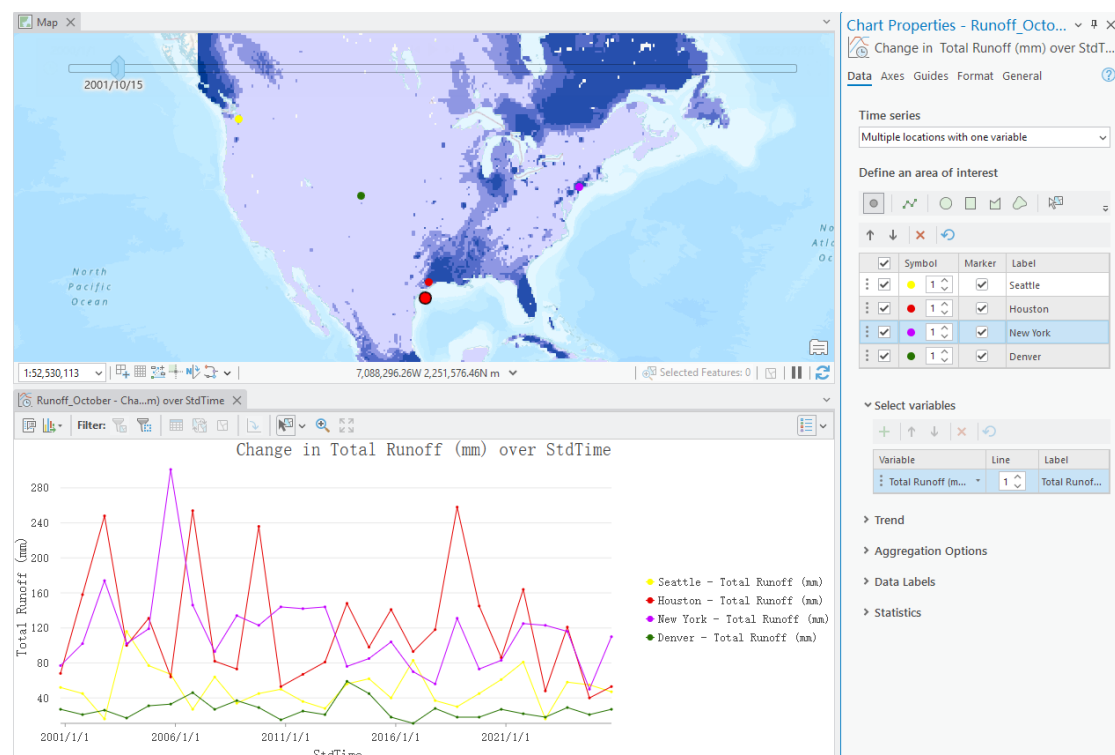


Q4

The average precipitation map shows that the eastern United States, the Gulf Coast, and parts of the Northwest Pacific have higher average precipitation. These regions receive a stable amount of moisture due to their proximity to the Atlantic, Gulf, and Pacific Oceans, resulting in abundant rainfall. Most of the southwestern United States and the western interior, including Nevada, Arizona, and parts of inland California, have lower average precipitation. These regions have arid or semi-arid climates, hence the relatively low average precipitation in October.

The standard deviation map shows that precipitation in the eastern United States and the Gulf Coast fluctuates significantly, with tropical storms and hurricanes causing substantial variations. Most of the western interior has a relatively low standard deviation for precipitation because these regions are relatively dry each year, resulting in relatively stable precipitation.

Q5



The time profile shows significant differences in October runoff at the four locations between 2000 and 2025. Houston and New York have high runoff and strong interannual variability; Seattle has moderate runoff; and Denver has the lowest runoff.

Seattle, located in the Pacific Northwest, receives precipitation influenced by Pacific moisture. While the region receives a relatively high annual precipitation, its October runoff is relatively moderate and stable compared to Houston and New York.

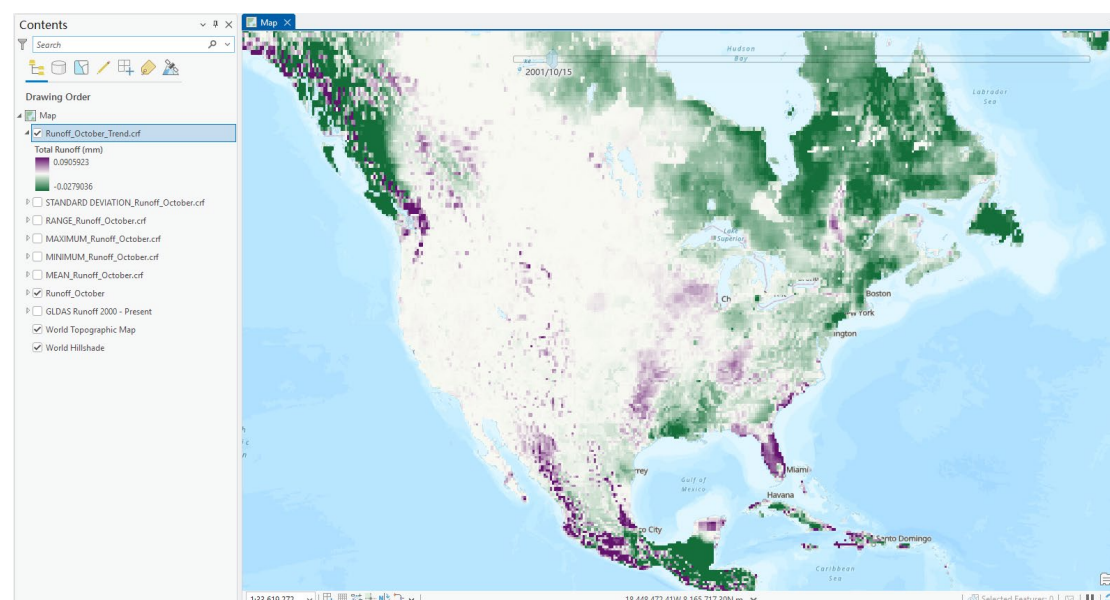
Denver has the lowest runoff among the four locations. This is likely due to the relatively limited precipitation and lower runoff resulting from the semi-arid climate of the Rocky Mountains interior.

Houston exhibits the greatest interannual variability in runoff, with multiple peaks exceeding 200 mm. This variability may be related to heavy rainfall events and tropical storm activity in the Gulf Coast region.

New York has relatively high runoff, but with moderate variability. Its humid continental climate and the influence of Atlantic moisture contribute to relatively stable October precipitation in the region.

Overall, coastal, humid regions such as Houston and New York tend to have higher and more variable runoff, while inland regions such as Denver have consistently lower runoff.

Q6



The slope map shows the long-term trend of total October runoff from 2000 to 2025. Positive slope values indicate that runoff increases over time, while negative slope

values indicate that runoff decreases over time. In parts of the eastern United States, such as New York and Boston, negative slope values indicate a downward trend in runoff during the study period, which may be due to changes in seasonal precipitation patterns. In many areas of the Midwestern United States, slope values remain unchanged or increase slightly, indicating that runoff in these areas remains stable with an upward trend, which may be related to an increase in heavy rainfall events.